

✓ Data VisualizationI

i)Use the in built dataset 'titanic'.The data set contains 891 rows and contains information about the passengers who boarded the unfortunate Titanic ship.Use the Seaborn library to see if we can find any patterns in the data.

ii)Write a codeto check howtheprice of the ticket (column name: 'fare')for each passenger is distributed byplottinga histogram.

iii)Use the in built dataset 'titanic'as used in the above problem. Plot a box plot for distribution of age with respect to each gender along with the information about whether they survived ornot. (column names: 'sex' and'age')

iv)Write observations on the inference from the abovestatistics.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
dataset= sns.load_dataset('titanic')
```

```
dataset.head()
```

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class
0	0	3	male	22.0	1	0	7.2500	S	Third
1	1	1	female	38.0	1	0	71.2833	C	First
2	1	3	female	26.0	0	0	7.9250	S	Third
3	1	1	female	35.0	1	0	53.1000	S	First
4	0	3	male	35.0	0	0	8.0500	S	Third

The data set contains 891 rows and 15 columns and contains information about the passengers who boarded the unfortunate Titanic ship. The original task is to predict whether or not the passenger survived depending upon different features such as their age, ticket, cabin they boarded, the class of the ticket, etc. We will use the Seaborn library to see if we can find any patterns in the data.

```
dataset.shape
```

```
(891, 15)
```

```
dataset.isnull()
```

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class
0	False	False	False	False	False	False	False	False	False
1	False	False	False	False	False	False	False	False	False
2	False	False	False	False	False	False	False	False	False
3	False	False	False	False	False	False	False	False	False
4	False	False	False	False	False	False	False	False	False
...	...	...	...	...	...	...	...	...	...
886	False	False	False	False	False	False	False	False	False
887	False	False	False	False	False	False	False	False	False
888	False	False	False	True	False	False	False	False	False
889	False	False	False	False	False	False	False	False	False
890	False	False	False	False	False	False	False	False	False

891 rows x 15 columns

```
dataset.isnull().sum()
```

	0
survived	0
pclass	0
sex	0
age	177
sibsp	0
parch	0
fare	0
embarked	2
class	0
who	0
adult_male	0
deck	688
embark_town	2
alive	0
alone	0

dtype: int64

```
# remove all null values from the dataset
dataset = dataset.dropna()
```

Distributional Plots

Distributional plots, as the name suggests are type of plots that show the statistical distribution of data.

The Dist Plot

The `distplot()` shows the histogram distribution of data for a single column. The column name is passed as a parameter to the `distplot()` function

```
# Let's see how the price of the ticket for each passenger is distributed
sns.distplot(dataset['fare'])
```

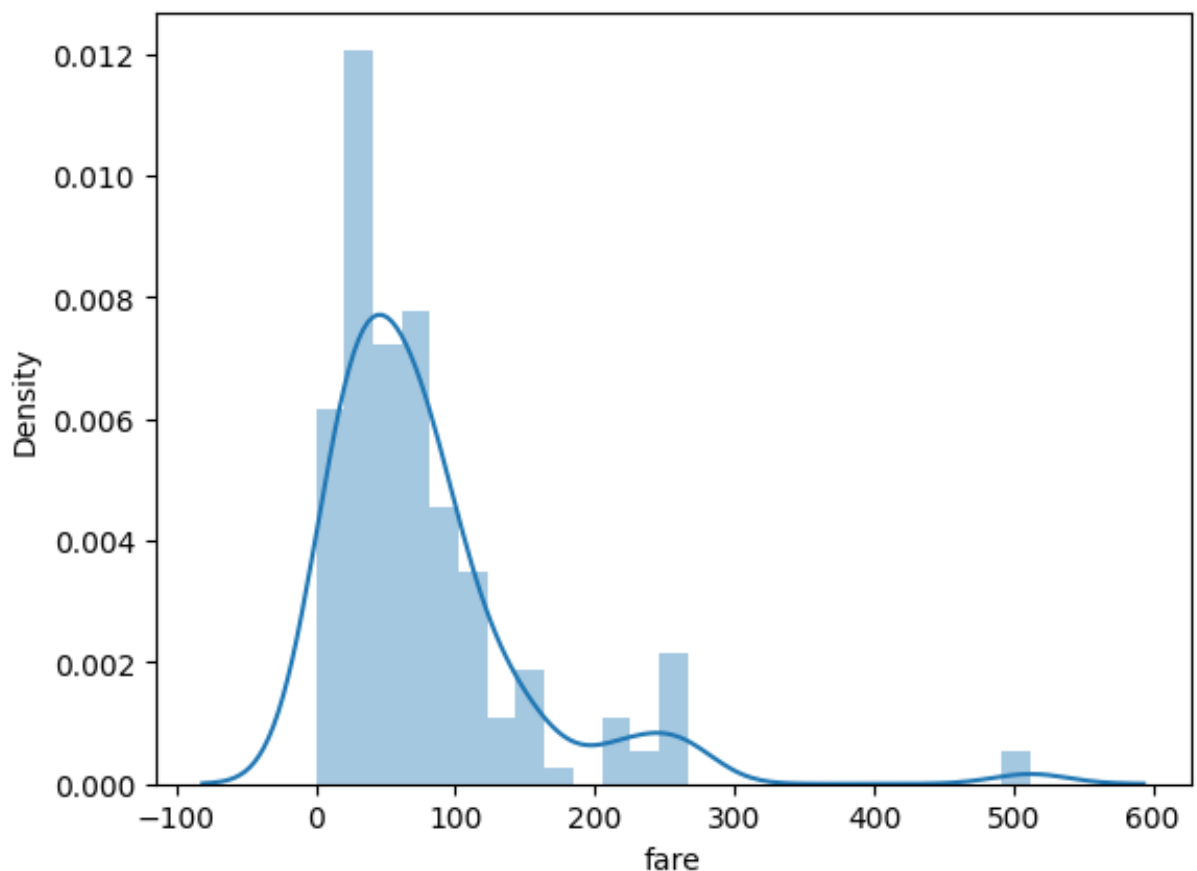
```
/tmp/ipykernel_15776/879315686.py:2: UserWarning:
```

```
`distplot` is a deprecated function and will be removed in seaborn v
```

```
Please adapt your code to use either `displot` (a figure-level function with
similar flexibility) or `histplot` (an axes-level function for histograms).
```

```
For a guide to updating your code to use the new functions, please see
https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751
```

```
sns.distplot(dataset['fare'])
<Axes: xlabel='fare', ylabel='Density'>
```



You can see that most of the ticket have been sold between 0-50 dollars. The line that you see represents the kernel density estimation. You can remove this line by passing `False` as the parameter for the `kde` attribute as shown below:

```
sns.distplot(dataset['fare'], kde=False)
```

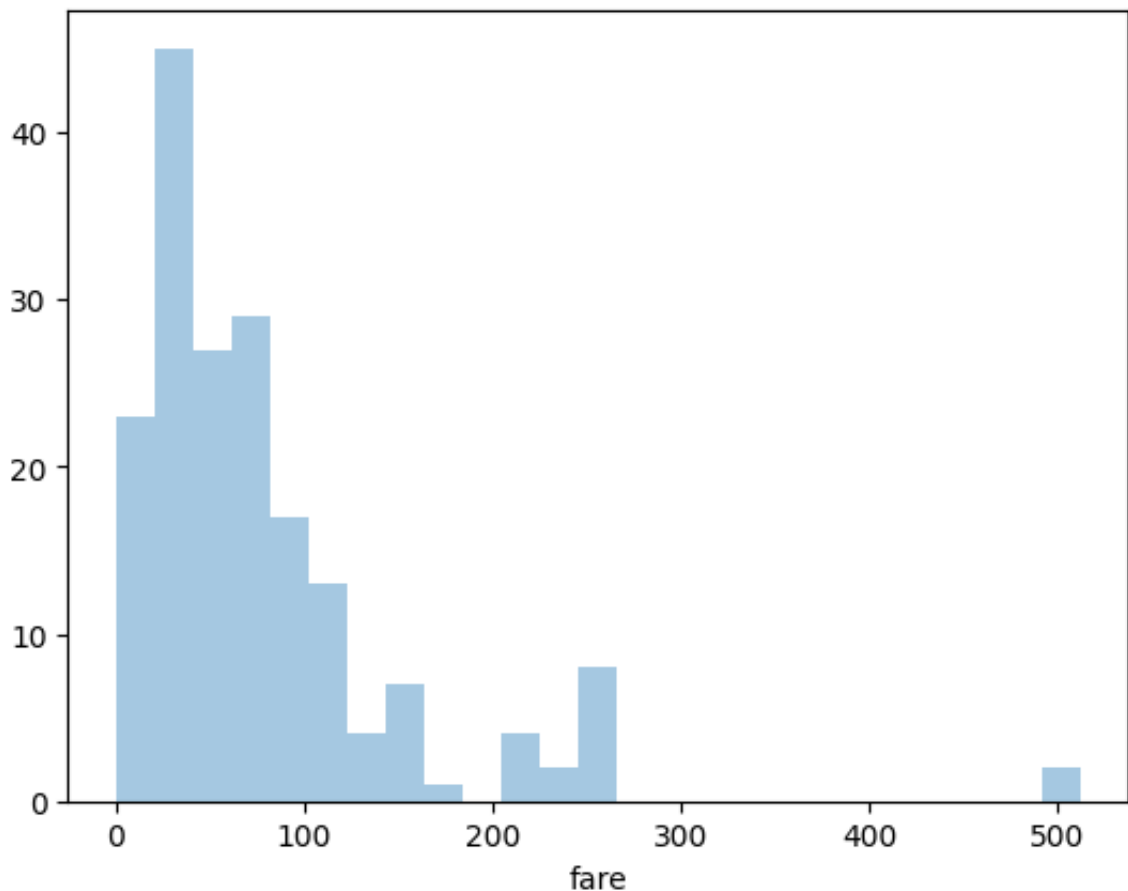
```
/tmp/ipykernel_15776/1623915515.py:1: UserWarning:
```

```
`distplot` is a deprecated function and will be removed in seaborn v
```

```
Please adapt your code to use either `displot` (a figure-level funct  
similar flexibility) or `histplot` (an axes-level function for histo
```

```
For a guide to updating your code to use the new functions, please s  
https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751
```

```
sns.distplot(dataset['fare'], kde=False)  
<Axes: xlabel='fare'>
```



Now you can see there is no line for the kernel density estimation on the plot. You can also pass the value for the bins parameter in order to see more or less details in the graph

```
sns.distplot(dataset['fare'], kde=False, bins=10) #Here we set the
```

```
/tmp/ipykernel_15776/2081720794.py:1: UserWarning:
```

```
`distplot` is a deprecated function and will be removed in seaborn v
```

Please adapt your code to use either `displot` (a figure-level function with similar flexibility) or `histplot` (an axes-level function for histograms).

For a guide to updating your code to use the new functions, please see <https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751>

```
sns.distplot(dataset['fare'], kde=False, bins=10) #Here we set the  
<Axes: xlabel='fare'>
```

